3.10 Formal Power Series and Laurent Series

Definition 3.10.1 (Formal Laurent Series)
*Let A be a commutative ring. Define the **formal Laurent series ring** $A((X))$ to be the set of sequences $(c_n)_{n\in\mathbb{Z}}$ in A such that there exists N for which $n \leq N \implies c_n = 0$. Addition is defined pointwise with product defined as follows*

$$\left(\sum_{n\in\mathbb{Z}} c_n X^n\right)\left(\sum_{n\in\mathbb{Z}} d_n X^n\right) = \sum_{n\in\mathbb{Z}}\left(\sum_{k\in\mathbb{Z}} c_{n-k}d_k\right)X^n$$

where the inner sum is a sum of a family with all but finitely many terms zero. That this is well-defined and makes $A((X))$ into a commutative ring is shown in (3.10.3).

Definition 3.10.2 (Order of a Formal Laurent Series)
*Let A be a commutative ring. For $0 \neq h = \sum_n c_n X^n \in A((X))$ define the **order** $v(h) := \min\{n \mid c_n \neq 0\}$, and $v(0) := \infty$. The minimum exists since the c_n vanish for n sufficiently negative and $h \neq 0$. We call $c_{v(h)}$ the **leading coefficient** of h.*

Proposition 3.10.3 ($A((X))$ is a Commutative Ring)
Let A be a commutative ring and $c, d \in A((X))$. For every $n \in \mathbb{Z}$ the family $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ has all but finitely many terms zero, so $(cd)_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ is defined, and $(cd)_n = 0$ for $n < v(c) + v(d)$. In particular $cd \in A((X))$. Moreover $A((X))$ is a commutative ring under pointwise addition and this product.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The product is well-defined. Let $c = (c_n)_{n\in\mathbb{Z}}$ and $d = (d_n)_{n\in\mathbb{Z}}$ be elements of $A((X))$. If $c = 0$ or $d = 0$ every term below is zero and the product is 0, so assume $v(c), v(d)$ are finite. Fix $n \in \mathbb{Z}$. If the term $c_{n-k}d_k$ is nonzero then $n - k \geq v(c)$ and $k \geq v(d)$, that is

$$v(d) \leq k \leq n - v(c)$$

so only finitely many $k \in \mathbb{Z}$ give a nonzero term. Hence $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ has all but finitely many terms zero, and the inner sum $e_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ is defined by (3.2.50). Moreover if $n < v(c) + v(d)$ then there is no integer k with $v(d) \leq k \leq n - v(c)$, so every term vanishes and $e_n = 0$. Therefore $(e_n)_{n\in\mathbb{Z}}$ is a sequence in A vanishing for $n < v(c) + v(d)$, that is an element of $A((X))$.

Ring axioms. Write $(cd)_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ for the coefficients of the product, an element of $A((X))$ by the first part.

Addition. For $c, d \in A((X))$ we have $c_n + d_n = 0$ and $-c_n = 0$ for $n < \min(v(c), v(d))$, so $c + d$ and $-c$ lie in $A((X))$, as does the zero sequence. So $A((X))$ is a subgroup of the abelian group $A^{\mathbb{Z}}$ under pointwise addition.

Identity. Let $1 := (\delta_{n,0})_{n\in\mathbb{Z}}$, which vanishes for $n \leq -1$ and so lies in $A((X))$. For $c \in A((X))$ the family $(\delta_{n-k,0}c_k)_{k\in\mathbb{Z}}$ has $k = n$ as its only possibly nonzero term, so $(1c)_n = c_n$.

Commutativity. The map $k \mapsto n - k$ is a bijection of $\mathbb{Z}$, so by (3.2.53) and commutativity of A,

$$(cd)_n = \sum_{k\in\mathbb{Z}} c_{n-k}d_k = \sum_{k\in\mathbb{Z}} c_k d_{n-k} = \sum_{k\in\mathbb{Z}} d_{n-k}c_k = (dc)_n$$

Distributivity. For $c, d, d' \in A((X))$, the families $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ and $(c_{n-k}d'_k)_{k\in\mathbb{Z}}$ have all but finitely many terms zero, so by (3.2.51)

$$(c(d + d'))_n = \sum_{k\in\mathbb{Z}} c_{n-k}(d_k + d'_k) = \sum_{k\in\mathbb{Z}} c_{n-k}d_k + \sum_{k\in\mathbb{Z}} c_{n-k}d'_k = (cd)_n + (cd')_n$$

and the other distributive law follows by commutativity.

Associativity. Let $c, d, e \in A((X))$ and fix $n \in \mathbb{Z}$; if any of them is 0 both sides are 0, so assume $v(c), v(d), v(e)$ are finite. Set $g(j,k) := c_{n-j-k}d_j e_k$ for $(j,k) \in \mathbb{Z}\times\mathbb{Z}$. If $g(j,k) \neq 0$ then $j \geq v(d)$, $k \geq v(e)$ and $n - j - k \geq v(c)$, so $v(d) \leq j \leq n - v(c) - v(e)$ and $v(e) \leq k \leq n - v(c) - v(d)$; hence g has all but finitely many terms zero. Recall that $(cd)_m = \sum_{j\in\mathbb{Z}} c_{m-j}d_j$ and $(de)_m = \sum_{k\in\mathbb{Z}} d_{m-k}e_k$. Then

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$$\begin{aligned}((cd)e)_n &= \sum_{k\in\mathbb{Z}} (cd)_{n-k}e_k && \text{by definition of the product}\\ &= \sum_{k\in\mathbb{Z}}\left(\sum_{j\in\mathbb{Z}} c_{n-k-j}d_j\right)e_k && \text{by definition of } (cd)_{n-k}\\ &= \sum_{k\in\mathbb{Z}}\sum_{j\in\mathbb{Z}} g(j,k) && \text{by (3.3.17)}\\ &= \sum_{(j,k)\in\mathbb{Z}\times\mathbb{Z}} g(j,k) && \text{by (3.2.53)}\end{aligned}$$

and on the other side

$$\begin{aligned}(c(de))_n &= \sum_{l\in\mathbb{Z}} c_{n-l}(de)_l && \text{by definition of the product}\\ &= \sum_{l\in\mathbb{Z}} c_{n-l}\sum_{k\in\mathbb{Z}} d_{l-k}e_k && \text{by definition of } (de)_l\\ &= \sum_{l\in\mathbb{Z}}\sum_{k\in\mathbb{Z}} g(l-k,k) && \text{by (3.3.17)}\\ &= \sum_{(l,k)\in\mathbb{Z}\times\mathbb{Z}} g(l-k,k) && \text{by (3.2.53)}\\ &= \sum_{(j,k)\in\mathbb{Z}\times\mathbb{Z}} g(j,k) && \text{by (3.2.53) for the bijection } (l,k)\mapsto(l-k,k)\end{aligned}$$

Hence $((cd)e)_n = (c(de))_n$ for all n. ∎

Lemma 3.10.4 (Valuation of a Sum and a Product)
Let A be a commutative ring and $c, d \in A((X))$. Then $v(1) = 0$, $v(-c) = v(c)$, and

a) $v(c+d) \geq \min(v(c), v(d))$
b) $v(cd) \geq v(c) + v(d)$
c) if A is an integral domain then so is $A((X))$, and $v(cd) = v(c) + v(d)$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Clearly $v(1) = 0$ and $v(-c) = v(c)$. If $n < \min(v(c), v(d))$ then $c_n = d_n = 0$, so $(c+d)_n = 0$, which gives a). For b), if $c = 0$ or $d = 0$ then $cd = 0$ and both sides are ∞. Otherwise $(cd)_n = 0$ for $n < v(c) + v(d)$ by (3.10.3), which means $v(cd) \geq v(c) + v(d)$. For c), let $c, d \neq 0$ with $m := v(c)$ and $r := v(d)$. A term $c_{m+r-k}d_k$ of $(cd)_{m+r}$ is nonzero only if $k \geq r$ and $m + r - k \geq m$, that is $k = r$, so $(cd)_{m+r} = c_m d_r \neq 0$ as A is a domain, while $(cd)_n = 0$ for $n < m + r$. Hence $cd \neq 0$ and $v(cd) = m + r$. ∎

Definition 3.10.5 (Formal Power Series)
*Let A be a commutative ring. Define the **formal power series ring** $A[[X]]$ to be the subring of $A((X))$ consisting of those sequences $(c_n)_{n\in\mathbb{Z}}$ for which $n < 0 \implies c_n = 0$, or equivalently those $h \in A((X))$ for which $v(h) \geq 0$.*

Proof that $A[[X]]$ is a subring of $A((X))$. Drafted by Claude Sonnet, reviewed by David Rufino.

$A[[X]] = \{h \in A((X)) \mid v(h) \geq 0\}$, so by (3.10.4) it contains 1 and is closed under negation, addition and multiplication. ∎

Definition 3.10.6 (Formal Derivative)
Let A be a commutative ring. For $f = \sum_n c_n X^n \in A((X))$, define $f' := \sum_n n c_n X^{n-1} \in A((X))$, that is $(f')_n = (n+1)c_{n+1}$.

Lemma 3.10.7 (Valuation of a Derivative)
Let A be a commutative ring and $f \in A((X))$. Then $v(f') \geq v(f) - 1$. In particular if $f \in A[[X]]$ then $f' \in A[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

If $n < v(f) - 1$ then $n + 1 < v(f)$, so $c_{n+1} = 0$ and $(f')_n = (n+1)c_{n+1} = 0$. This gives $v(f') \geq v(f) - 1$. If $f \in A[[X]]$ then $v(f) \geq 0$ and $(f')_n = 0$ for $n < -1$; also $(f')_{-1} = 0 \cdot c_0 = 0$, so $f' \in A[[X]]$. ∎

Definition 3.10.8 (Summable Family in $A((X))$)
*Let A be a commutative ring. A family $(h_i)_{i\in I}$ in $A((X))$ is **summable** if for every $N \in \mathbb{Z}$ all but finitely many $i \in I$ satisfy $v(h_i) \geq N$. Its **sum** is the formal Laurent series*

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$$\sum_{i\in I} h_i := \sum_{n\in\mathbb{Z}} \left(\sum_{i\in I} (h_i)_n \right) X^n$$

where $(h_i)_n$ denotes the coefficient of X^n in h_i.

Proof that the sum is well-defined. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $h_{i,n} := (h_i)_n$ and fix $n \in \mathbb{Z}$. By hypothesis (with $N = n+1$) the set $\{i \in I \mid v(h_i) \le n\}$ is finite, and it contains every i with $h_{i,n} \neq 0$. So $(h_{i,n})_{i\in I}$ has all but finitely many terms zero and $s_n := \sum_{i\in I} h_{i,n}$ is defined by (3.2.50). Let $J_0 := \{i \in I \mid v(h_i) \le 0\}$, which is finite, and $N_0 := \min\big(0, \min_{i\in J_0} v(h_i)\big)$ (with $N_0 := 0$ if J_0 is empty). If $n < N_0$ then $v(h_i) > n$ for every i, so every $h_{i,n} = 0$ and $s_n = 0$. Hence $(s_n)_{n\in\mathbb{Z}}$ lies in $A((X))$. ∎

Proposition 3.10.9 (Sums of Summable Families)
Let A be a commutative ring and $(h_i)_{i\in I}$ a summable family in $A((X))$. Then

a) If $v(h_i) \ge N$ for every $i \in I$ then $v\big(\sum_{i\in I} h_i\big) \ge N$. In particular if every $h_i \in A[[X]]$ then $\sum_{i\in I} h_i \in A[[X]]$.
b) For a bijection $\sigma : I' \to I$ the family $\big(h_{\sigma(i')}\big)_{i'\in I'}$ is summable and $\sum_{i'\in I'} h_{\sigma(i')} = \sum_{i\in I} h_i$.
c) The family $(h'_i)_{i\in I}$ is summable and $\big(\sum_{i\in I} h_i\big)' = \sum_{i\in I} h'_i$.
d) For any $g \in A((X))$ the family $(gh_i)_{i\in I}$ is summable and $g \cdot \sum_{i\in I} h_i = \sum_{i\in I} gh_i$.
e) If $(k_i)_{i\in I}$ is another summable family then $(h_i + k_i)_{i\in I}$ is summable and $\sum_{i\in I}(h_i + k_i) = \sum_{i\in I} h_i + \sum_{i\in I} k_i$.
f) If $I = \bigsqcup_{\lambda\in\Lambda} I_\lambda$ is a partition of I into disjoint subsets then each $(h_i)_{i\in I_\lambda}$ and $\big(\sum_{i\in I_\lambda} h_i\big)_{\lambda\in\Lambda}$ is summable, and $\sum_{i\in I} h_i = \sum_{\lambda\in\Lambda} \sum_{i\in I_\lambda} h_i$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $h_{i,n} := (h_i)_n$ and $s_n := \sum_{i\in I} h_{i,n} = \big(\sum_{i\in I} h_i\big)_n$.

a) If $n < N$ then $h_{i,n} = 0$ for every i, so $s_n = 0$, as $J = \emptyset$ is a valid choice in (3.2.50). Hence $v\big(\sum_{i\in I} h_i\big) \ge N$, and taking $N = 0$ gives the statement for $A[[X]]$.

b) The reindexed family clearly meets the summability condition. The family $(h_i)_{i\in I}$ itself is not almost zero in $A((X))$, but for each n the coefficient family $(h_{i,n})_{i\in I}$ in A is, so the statement holds coefficientwise by (3.2.53).

c) Since $v(h'_i) \ge v(h_i) - 1$ by (3.10.7) the family $(h'_i)_{i\in I}$ is also summable, so $\sum_{i\in I} h'_i$ is defined. Using $(h')_n = (n+1)h_{n+1}$, for every $n \in \mathbb{Z}$

$$\begin{aligned}
\left(\left(\sum_{i\in I} h_i \right)' \right)_n &= (n+1)s_{n+1} = (n+1)\sum_{i\in I} h_{i,n+1} && \text{by definition of the sum} \\
&= \sum_{i\in I} (n+1)h_{i,n+1} && \text{by (3.3.17)} \\
&= \sum_{i\in I} (h'_i)_n = \left(\sum_{i\in I} h'_i \right)_n && \text{by definition of the sum}
\end{aligned}$$

d) Let $g \in A((X))$. If $g = 0$ both sides are 0, so assume $v(g)$ is finite. By (3.10.4) $v(gh_i) \ge v(g) + v(h_i)$, so $(gh_i)_{i\in I}$ is summable and $\sum_{i\in I} gh_i$ is defined. Fix $n \in \mathbb{Z}$ and set $a(k,i) := g_{n-k}h_{i,k}$ for $(k,i) \in \mathbb{Z} \times I$. If $a(k,i) \neq 0$ then $k \le n - v(g)$ and $v(h_i) \le k$, so i lies in the finite set $\{i \in I \mid v(h_i) \le n - v(g)\}$ and $v(h_i) \le k \le n - v(g)$; hence a has all but finitely many terms zero. Then

$$\begin{aligned}
\left(g \sum_{i\in I} h_i \right)_n &= \sum_{k\in\mathbb{Z}} g_{n-k}s_k = \sum_{k\in\mathbb{Z}} g_{n-k} \sum_{i\in I} h_{i,k} && \text{by definition of the product and the sum} \\
&= \sum_{k\in\mathbb{Z}} \sum_{i\in I} a(k,i) && \text{by (3.3.17)} \\
&= \sum_{i\in I} \sum_{k\in\mathbb{Z}} a(k,i) && \text{by (3.2.53)} \\
&= \sum_{i\in I} \sum_{k\in\mathbb{Z}} g_{n-k}h_{i,k} && \text{by definition of } a \\
&= \sum_{i\in I} (gh_i)_n = \left(\sum_{i\in I} gh_i \right)_n && \text{by definition of the product and the sum}
\end{aligned}$$

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e) By (3.10.4) a), $v(h_i+k_i)\geq \min(v(h_i),v(k_i))$, so for every N all but finitely many i satisfy $v(h_i+k_i)\geq N$ and $(h_i+k_i)_{i\in I}$ is summable. The identity holds in each coefficient by (3.2.51).

f) The subfamilies are summable. For $N\in\mathbb{Z}$ the set $F_N := \{i\in I \mid v(h_i)<N\}$ is finite, so only finitely many λ have $I_\lambda\cap F_N\neq\emptyset$, and for every other λ we have $v\left(\sum_{i\in I_\lambda} h_i\right)\geq N$ by a). Hence $\left(\sum_{i\in I_\lambda} h_i\right)_{\lambda\in\Lambda}$ is summable. The identity holds in each coefficient by (3.2.52) applied to $\left(h_{i,n}\right)_{i\in I}$. ∎

Proposition 3.10.10 (Units of a Formal Power Series Ring)
Let A be a commutative ring and $u\in A[[X]]$, with constant coefficient u_0 (the coefficient of X^0). Then u is a unit of $A[[X]]$ if and only if $u_0\in A^\star$. In particular, when $A=k$ is a field, $u\in k[[X]]^\star \iff u_0\neq 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose $u_0\in A^\star$ and set $w := 1-u_0^{-1}u\in A[[X]]$. Its constant coefficient is 0, that is $v(w)\geq 1$, so $v(w^i)\geq i$ for all $i\geq 0$ by (3.10.4). Hence $(w^i)_{i\geq 0}$ is summable (3.10.8) and $s := \sum_{i\geq 0} w^i\in A[[X]]$ by (3.10.9). Then

$$\begin{aligned} s &= 1+\sum_{i\geq 1} w^i && \text{splitting off } i=0 \text{ by (3.10.9) } f)\\ &= 1+\sum_{i\geq 0} w^{i+1} && \text{reindexing } i\mapsto i+1 \text{ by (3.10.9) } b)\\ &= 1+ws && \text{distributivity, by (3.10.9) } d)\end{aligned}$$

so $(1-w)s=1$. Since $u_0^{-1}u=1-w$, this says $u\cdot(u_0^{-1}s)=1$, so $u_0^{-1}s$ is an inverse of u.

Conversely, if u is a unit with inverse $t\in A[[X]]$ then $(ut)_0=\sum_{k\in\mathbb{Z}} u_{-k}t_k = u_0t_0$, as $u_{-k}=0$ for $k>0$ and $t_k=0$ for $k<0$. So $u_0t_0=1$ and $u_0\in A^\star$. When $A=k$ is a field, $u_0\in k^\star$ is equivalent to $u_0\neq 0$. ∎

Proposition 3.10.11 (Units of a Formal Laurent Series Ring)
Let A be an integral domain and $0\neq h\in A((X))$ with $m := v(h)$. The following are equivalent

a) h is a unit of $A((X))$
b) $h=X^m u$ for a unit $u\in A[[X]]^\star$
c) the leading coefficient h_m of h is a unit of A

In particular when $A=k$ is a field, $k((X))$ is a field.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

c) $\Longrightarrow$ b): $u := X^{-m}h\in A[[X]]$ has constant coefficient $u_0=h_m\in A^\star$, so u is a unit of $A[[X]]$ by (3.10.10).

b) $\Longrightarrow$ a): $X^{-m}u^{-1}$ is an inverse of $h=X^m u$.

a) $\Longrightarrow$ c): let $hh'=1$ in $A((X))$. Then $v(h)+v(h')=v(1)=0$ by (3.10.4) c), so $v(h')=-m$, and the coefficient of X^0 in hh' is $h_mh'_{-m}=1$, so $h_m\in A^\star$. ∎

Proposition 3.10.12 ($k((X))$ is a Field of Fractions for $k[[X]]$)
Let k be a field. Then every element of the field $k((X))$ is a/b for $a,b\in k[[X]]$, $b\neq 0$ – that is, $k((X))$ is a field of fractions for the subring $k[[X]]\subset k((X))$. Moreover, the order function $v: k((X))^\star\to\mathbb{Z}$ is a group homomorphism under multiplication: $v(fg)=v(f)+v(g)$, and $v(f^{-1})=-v(f)$.

Proof. By (3.10.11), as k is a field every $c\neq 0$ in $k((X))$ is $c=X^m u$ with $m=v(c)$ and $u\in k[[X]]^\star$, which is $uX^m\in k[[X]]$ if $m\geq 0$ and u/X^{-m} otherwise, in either case a ratio of two elements of $k[[X]]$.

By (3.10.4) c), as k is a domain, $v(fg)=v(f)+v(g)$ for $f,g\neq 0$. Applying this to f,f^{-1}: $0=v(1)=v(ff^{-1})=v(f)+v(f^{-1})$, giving $v(f^{-1})=-v(f)$. ∎

Proposition 3.10.13 (Formal Power Series Ring is a DVR)
Let k be a field. Then $k[[X]]$ is an integral domain, not a field, in which every nonzero element a can be written uniquely as $a=X^{v(a)}u$ with $u\in k[[X]]^\star$ (equivalently, $u(0)\neq 0$ (3.10.10)).

Using the criteria (3.31.2)e) $k[[X]]$ is a discrete valuation ring with valuation precisely the order function v.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$k[[X]]$ is a domain, as a subring of the domain $k((X))$ ((3.10.4) c)), and not a field, as $X\neq 0$ has no inverse ($v(Xg)=1+v(g)\neq 0$ for every $g\in k[[X]]$ by the same result). Given $0\neq a\in k[[X]]$, by (3.10.11) $a=X^{v(a)}u$ with $u\in k[[X]]^\star$, where $v(a)\geq 0$: this is exactly the uniformised-element criterion (3.31.2)e), with $\pi := X$. If also $a=X^n u'$ with $u'\in$

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$k[[X]]^\star$ then $v(u')=0$ and $v(a)=n+v(u')=n$ by (3.10.4) c), so the exponent is $v(a)$ and then $u'=u$ as $k((X))$ is a domain. ∎

Proposition 3.10.14 (Product Rule)
Let A be a commutative ring and $f,g\in A((X))$. Then $(fg)'=f'g+fg'$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $f=\sum_i a_iX^i$, $g=\sum_j b_jX^j$ and fix $n\in\mathbb{Z}$. All sums below are over $i\in\mathbb{Z}$ and are sums of families with all but finitely many terms zero, with the coefficients of a product as in (3.10.3). The coefficient of X^{n-1} in $(fg)'$ is $n(fg)_n=n\sum_i a_ib_{n-i}$. Since $(f')_k=(k+1)a_{k+1}$ and $(g')_k=(k+1)b_{k+1}$, reindexing $i=k+1$ and $i=n-k-1$ respectively ((3.2.53)) gives

$$(f'g)_{n-1}=\sum_i ia_ib_{n-i}\qquad (fg')_{n-1}=\sum_i(n-i)a_ib_{n-i}$$

Hence

$$\begin{aligned}(f'g+fg')_{n-1}&=\sum_i ia_ib_{n-i}+\sum_i(n-i)a_ib_{n-i} && \text{by the above}\\ &=\sum_i na_ib_{n-i} && \text{by (3.2.51)}\\ &=n\sum_i a_ib_{n-i}=((fg)')_{n-1} && \text{by (3.3.17)}\end{aligned}$$

As n was arbitrary, $(fg)'=f'g+fg'$. ∎

Lemma 3.10.15 (Power Rule)
Let A be a commutative ring and $\psi\in A((X))$. Then $(\psi^n)'=n\psi^{n-1}\psi'$ for every $n\geq 0$. If moreover ψ is a unit of $A((X))$, this also holds for every $n<0$.

Proof. **$n\geq 0$.** Induction on n: the case $n=0$ is $0=0$ ($\psi^0=1$ is constant). Given $(\psi^n)'=n\psi^{n-1}\psi'$, the product rule (3.10.14) gives

$$(\psi^{n+1})'=(\psi\cdot\psi^n)'=\psi'\psi^n+\psi(\psi^n)'=\psi'\psi^n+n\psi^n\psi'=(n+1)\psi^n\psi'$$

$n<0$, ψ a unit. Differentiating $\psi\psi^{-1}=1$ with the product rule, $0=\psi'\psi^{-1}+\psi(\psi^{-1})'$, so $(\psi^{-1})'=-\psi^{-2}\psi'$ – the case $n=-1$. For $n\leq -2$, write $n=-r$, $r\geq 2$: applying the $n\geq 0$ case just shown to the element $\psi^{-1}\in A((X))$ (no hypothesis on its order was used there),

$$(\psi^{-r})'=\left((\psi^{-1})^r\right)'=r(\psi^{-1})^{r-1}(\psi^{-1})'=r(\psi^{-1})^{r-1}(-\psi^{-2}\psi')=-r\psi^{-r-1}\psi'=n\psi^{n-1}\psi'$$

∎

Proposition 3.10.16 (Substitution into a Formal Laurent Series)
Let A be an integral domain, $f=\sum_{n\geq n_0}c_nX^n\in A((X))$ and $\psi\in XA[[X]]$ (that is, $\psi(0)=0$). Suppose that either

- *$f\in A[[X]]$, that is $n_0\geq 0$, or*
- *$\psi\in A((X))^\star$, that is $\psi\in X^mA[[X]]^\star$ for some $m>0$ (3.10.11)*

Then

$$(f\circ\psi)(X):=\sum_{n\geq n_0}c_n\psi(X)^n$$

is a well-defined element of $A((X))$ (with ψ^n, $n<0$, the power in the ring $A((X))$), and $f\circ\psi\in A[[X]]$ if $f\in A[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\psi\in XA[[X]]$ has no terms of degree ≤ 0 we have $v(\psi)\geq 1$, so $v(\psi^n)\geq n$ for every $n\geq 0$ by (3.10.4) b). Negative powers of ψ only occur for the finitely many n with $n_0\leq n<0$. Hence for every N all $n\geq\max(N,0)$ satisfy $v(c_n\psi^n)\geq n\geq N$, and only the finitely many n with $n_0\leq n<\max(N,0)$ can fail this, so $(c_n\psi^n)_{n\geq n_0}$ is summable (3.10.8) and its sum lies in $A((X))$. If $f\in A[[X]]$ then every $c_n\psi^n\in A[[X]]$ and the sum lies in $A[[X]]$ by (3.10.9) a). ∎

Proposition 3.10.17 (Chain Rule)
In the setting of (3.10.16), $(f\circ\psi)'=(f'\circ\psi)\cdot\psi'$.

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Proof. By the power rule ((3.10.15), which for $n < 0$ requires that ψ is a unit of $A((X))$, as it is then), $(c_n\psi^n)' = c_n(\psi^n)' = nc_n\psi^{n-1}\psi'$ for every $n \geq n_0$, using linearity of $(-)'$ (immediate from (3.10.6), as $c_n \in A$ is a constant). The term $n = 0$ is 0, and is simply omitted if ψ is not a unit. Then, by termwise differentiation (3.10.9) c) of the summable family $(c_n\psi^n)_{n\geq n_0}$ ((3.10.16)'s proof),

$$(f \circ \psi)' = \sum_{n\geq n_0} (c_n\psi^n)' = \sum_{n\geq n_0} nc_n\psi^{n-1}\psi'$$

The family $(nc_n\psi^{n-1})_{n\geq n_0}$ is summable by the same order bound as $(c_n\psi^n)_{n\geq n_0}$ ((3.10.16)'s proof, shifted down one degree), so distributivity ((3.10.9) d)) gives

$$\sum_{n\geq n_0} nc_n\psi^{n-1}\psi' = \left(\sum_{n\geq n_0} nc_n\psi^{n-1}\right)\psi'$$

Reindexing $m = n - 1$, $\sum_{n\geq n_0} nc_n\psi^{n-1} = \sum_{m\geq n_0-1}(m+1)c_{m+1}\psi^m = f' \circ \psi$, giving $(f \circ \psi)' = (f' \circ \psi) \cdot \psi'$. ∎

Proposition 3.10.18 (Substitution is a Ring Homomorphism)
Let A be an integral domain and $\psi \in XA[[X]]$. Then $f \mapsto f \circ \psi$ is a ring homomorphism $A[[X]] \to A[[X]]$. If moreover the leading coefficient of ψ is a unit (so $\psi \in A((X))^\star$ by (3.10.11)) then $f \mapsto f \circ \psi$ is a ring homomorphism $A((X)) \to A((X))$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The arguments below apply to f, g, h in $A[[X]]$ in the first case and in $A((X))$ in the second.

Identity. $1 \circ \psi = \psi^0 = 1$.

Additivity. Write $f = \sum_{n\geq n_0} c_nX^n$, $h = \sum_{n\geq m_0} d_nX^n$, and set $p_0 := \min(n_0, m_0)$, extending both coefficient sequences by $c_n := 0$ (resp. $d_n := 0$) for $p_0 \leq n < n_0$ (resp. $< m_0$); this changes neither series, nor the sums below, as it only adds zero terms. Then $f + h = \sum_{n\geq p_0}(c_n + d_n)X^n$, and the families $(c_n\psi^n)_{n\geq p_0}$ and $(d_n\psi^n)_{n\geq p_0}$ are summable by (3.10.16). Hence

$$\begin{aligned}(f+h)\circ\psi &= \sum_{n\geq p_0}(c_n+d_n)\psi^n = \sum_{n\geq p_0}(c_n\psi^n + d_n\psi^n) && \text{by distributivity in } A((X))\\ &= \sum_{n\geq p_0} c_n\psi^n + \sum_{n\geq p_0} d_n\psi^n = f\circ\psi + h\circ\psi && \text{by (3.10.9) } e)\end{aligned}$$

Multiplicativity. Write $f = \sum_{i\geq i_0} c_iX^i$, $g = \sum_{j\geq j_0} d_jX^j$, so $f \circ \psi = \sum_i c_i\psi^i$ and $g \circ \psi = \sum_j d_j\psi^j$, and let $P := \mathbb{Z}_{\geq i_0} \times \mathbb{Z}_{\geq j_0}$ and $a(i,j) := c_id_j\psi^{i+j}$ for $(i,j) \in P$. We have $v(a(i,j)) \geq i + j$ whenever $i + j \geq 0$ as $v(\psi^n) \geq n$ for $n \geq 0$ ((3.10.16)'s proof), and for every N only the finitely many pairs with $i + j < \max(N, 0)$ can fail $v(a(i,j)) \geq N$, so $(a(i,j))_{(i,j)\in P}$ is summable. Then

$$\begin{aligned}(f\circ\psi)(g\circ\psi) &= \sum_{j\geq j_0}(f\circ\psi)d_j\psi^j && \text{by (3.10.9) } d)\\ &= \sum_{j\geq j_0}\sum_{i\geq i_0} c_id_j\psi^{i+j} && \text{by (3.10.9) } d)\\ &= \sum_{(i,j)\in P} a(i,j) && \text{by (3.10.9) } f) \text{ for the partition of } P \text{ by } j\end{aligned}$$

and, grouping instead by $N := i + j$, whose fibres are finite,

$$\begin{aligned}(fg)\circ\psi &= \sum_N\left(\sum_{i+j=N} c_id_j\right)\psi^N = \sum_N\sum_{i+j=N} c_id_j\psi^N && \text{by (3.3.17)}\\ &= \sum_{(i,j)\in P} a(i,j) && \text{by (3.10.9) } f) \text{ for the partition of } P \text{ by } i+j\end{aligned}$$

so $(f \circ \psi)(g \circ \psi) = (fg) \circ \psi$. ∎

3.10.1 Exponential and Logarithm

Definition 3.10.19 (Formal Exponential and Logarithm)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Define

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$$\exp(X) \coloneqq \sum_{n\geq 0} \frac{X^n}{n!} \in A[[X]] \qquad \log(1+X) \coloneqq \sum_{n\geq 1} (-1)^{n+1} \frac{X^n}{n} \in XA[[X]]$$

For $f \in XA[[X]]$ and $g \in 1+XA[[X]]$ define $\exp(f) \coloneqq \exp(X) \circ f$ and $\log(g) \coloneqq \log(1+X) \circ (g-1)$, using substitution (3.10.16) (note $g-1 \in XA[[X]]$).

Lemma 3.10.20 (Zero Derivative)
Let A be a commutative ring with $\mathbb{Q} \subseteq A$ and $F \in A((X))$ with $F' = 0$. Then F is a constant, that is $F \in A$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$(F')_n = (n+1)F_{n+1} = 0$ for every $n \in \mathbb{Z}$, and $n+1$ is invertible in A for $n \neq -1$, so $F_{n+1} = 0$ for all $n \neq -1$, that is $F = F_0$. ∎

Proposition 3.10.21 (Derivatives of exp and log)
Let A be an integral domain with $\mathbb{Q} \subseteq A$, $f \in XA[[X]]$ and $g \in 1+XA[[X]]$. Then $\exp(f) \in 1+XA[[X]]$ and $\log(g) \in XA[[X]]$, and

$$\exp(f)' = \exp(f)f' \qquad \log(g)' = g'g^{-1}$$

where g is a unit of $A[[X]]$ by (3.10.10).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

exp. Directly $\exp(X)' = \sum_{n\geq 1} \frac{X^{n-1}}{(n-1)!} = \exp(X)$. Since $v(f^n) \geq n$, only the term $n=0$ contributes to the constant coefficient of $\exp(f) = \sum_n \frac{f^n}{n!}$, so $\exp(f) \in 1+XA[[X]]$. By the chain rule (3.10.17) $\exp(f)' = (\exp' \circ f)f' = \exp(f)f'$.

log. Likewise $\log(g) \in XA[[X]]$, as each term of $\sum_{n\geq 1} (-1)^{n+1} \frac{(g-1)^n}{n}$ has order at least n. Directly $\log(1+X)' = \sum_{n\geq 1} (-1)^{n+1} X^{n-1} = \sum_{m\geq 0} (-X)^m = (1+X)^{-1}$, as $(1+X)\sum_{m\geq 0}(-X)^m = 1$ coefficientwise. By the chain rule

$$\log(g)' = (\log(1+X)' \circ (g-1))g' = ((1+X)^{-1} \circ (g-1))g' = g^{-1}g'$$

since $f \mapsto f \circ (g-1)$ is a ring homomorphism (3.10.18) and $(1+X) \circ (g-1) = g$. ∎

Proposition 3.10.22 (Functional Equations of exp and log)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Then $\exp(f+g) = \exp(f)\exp(g)$ for $f, g \in XA[[X]]$, and $\log(gh) = \log(g) + \log(h)$ for $g, h \in 1+XA[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

exp. Write $E \coloneqq \exp(f+g)$ and $U \coloneqq \exp(f)\exp(g)$, a unit of $A[[X]]$ with constant coefficient 1. By the product rule and (3.10.21), $E' = E(f'+g')$ and $U' = U(f'+g')$. Hence for $R \coloneqq EU^{-1}$, using the power rule (3.10.15) for U^{-1},

$$\begin{aligned} R' &= E'U^{-1} - EU'U^{-2} && \text{by the product and power rules} \\ &= E(f'+g')U^{-1} - EU(f'+g')U^{-2} = 0 \end{aligned}$$

so R is constant by (3.10.20), equal to $R(0) = 1$, that is $E = U$.

log. By the product rule and (3.10.21), $\log(gh)' = (g'h + gh')g^{-1}h^{-1} = g'g^{-1} + h'h^{-1} = (\log(g) + \log(h))'$. So $\log(gh) - \log(g) - \log(h)$ has derivative 0 and constant coefficient 0, hence is 0 by (3.10.20). ∎

Proposition 3.10.23 (exp and log are Inverse Isomorphisms)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Then $\log(\exp(f)) = f$ for $f \in XA[[X]]$ and $\exp(\log(g)) = g$ for $g \in 1+XA[[X]]$. Hence exp and log are mutually inverse isomorphisms of groups between $(XA[[X]], +)$ and $(1+XA[[X]], \times)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.10.21) $\log(\exp(f))' = \exp(f)'\exp(f)^{-1} = f'$, so $\log(\exp(f)) - f$ has derivative 0 and constant coefficient 0, and is 0 by (3.10.20). For the other composite let $E \coloneqq \exp(\log(g))$, so $E' = E\log(g)' = Eg'g^{-1}$ by the same result. Then

$$\begin{aligned} (Eg^{-1})' &= E'g^{-1} - Eg'g^{-2} && \text{by the product and power rules} \\ &= Eg'g^{-2} - Eg'g^{-2} = 0 \end{aligned}$$

so Eg^{-1} is constant by (3.10.20), equal to $E(0)g(0)^{-1} = 1$, that is $E = g$. Finally exp is a homomorphism by (3.10.22), with inverse log. ∎

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